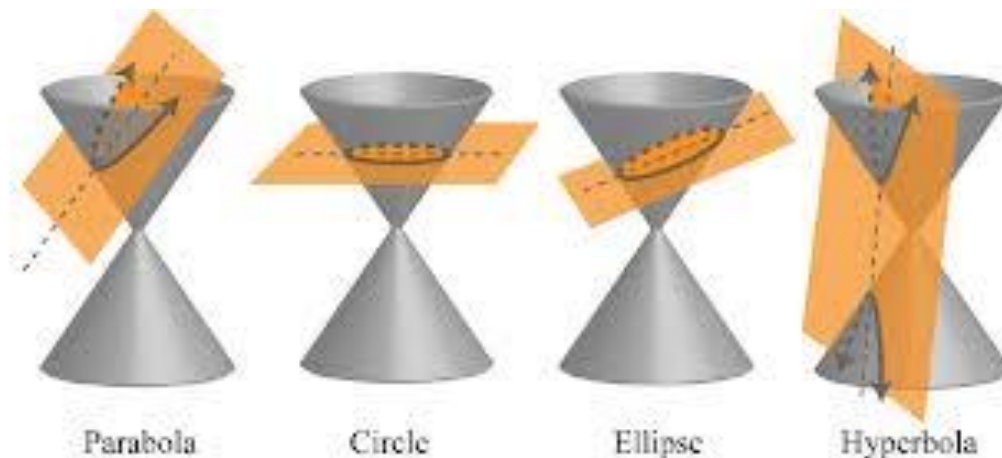


5.4.1 Conic Sections (Parabolas)

If a plane passes through a double right circular cone there are four possible resulting curves. These curves are the **parabola**, **circle**, **ellipse**, and **hyperbola**. These curves are called **conic sections** and are shown below.



Definition: A **parabola** is the set of all points in the plane that are equidistant from a fixed line (the **directrix**) and a fixed point not on the line (the **focus**)

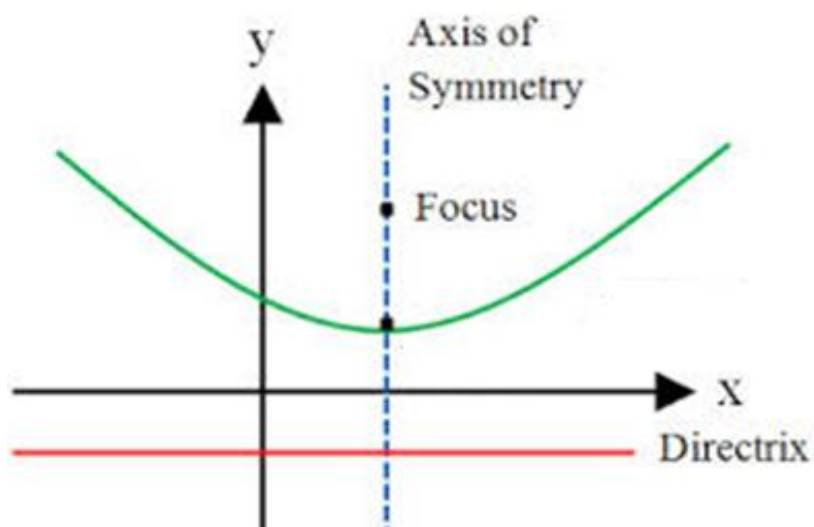
The Equation of a Parabola: (opening up or down)

The parabola

has focus

and directrix

Note: If $a > 0$ then the parabola opens up. If $a < 0$ then the parabola opens down.



Example 1: Find the equation of the parabola with focus $(-1, 3)$ and directrix $y = 2$.

Example 2: Find the vertex, focus, and directrix of the graph of $y = -3x^2 - 6x + 2$.

Example 3: Find the vertex, focus, and directrix of the graph of $y = 2x^2 + 6x - 7$.

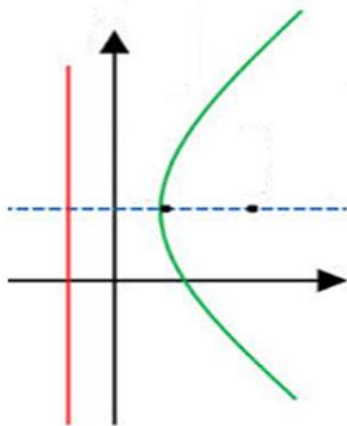
The Equation of a Parabola: (opening left or right)

The parabola

has focus

and directrix

Note: If $a > 0$ then the parabola opens to the right. If $a < 0$ then the parabola opens to the left..



Example 4: Find the vertex, axis of symmetry, y-intercepts, focus, and directrix for the parabola $x = y^2 - 2y$. Sketch a graph of this parabola.